RIT | Golisano College of Computing and Information Sciences
School of Information

ISTE-230 Introduction to Database & Data Modeling
Practice Exercise # 11 – Relational Algebra Statements

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All assignments will be graded for correctness and the use of the standards that were discussed in class and are documented in the Standards Content area.

PART I

For each question, show the theoretical relational algebra statement that would accomplish what is requested (not SQL statements) and the resulting relation.

Use the following relations to answer the questions below:

STUDENT

studentID	name	major
123	Bill	IT
234	Sue	CS
345	Tom	SE
456	Ann	BUS
567	Linda	IT
678	Tom	IT
789	Sue	LA

ITSTUDENT

studentID	name	major
123	Bill	IT
567	Linda	IT
678	Tom	IT
890	Jon	IT
901	Lynn	IT

1. Perform a union of STUDENT and ITSTUDENT?

   STATEMENT: `STUDENT UNION ITSTUDENT`

   RELATION:

studentID	name	major
123	Bill	IT
234	Sue	CS
345	Tom	SE
456	Ann	BUS
567	Linda	IT
678	Tom	IT
789	Sue	LA
890	Jon	IT
901	Lynn	IT

2. Perform an intersection of STUDENT and ITSTUDENT?

STATEMENT: `STUDENT INTERSECT ITSTUDENT`

RELATION:

studentID	name	major
123	Bill	IT
567	Linda	IT
678	Tom	IT

3. Perform a difference of STUDENT and ITSTUDENT?

STATEMENT: `STUDENT DIFFERENCE ITSTUDENT`

RELATION:

studentID	name	major
234	Sue	CS
345	Tom	SE
456	Ann	BUS
789	Sue	LA

4. Perform a difference of ITSTUDENT and STUDENT?

STATEMENT: `ITSTUDENT DIFFERENCE STUDENT`

RELATION:

studentID	Name	Major
890	Jon	IT
901	Lynn	IT

PART II.

For each question, show the theoretical relational algebra statement that would accomplish what is requested (not SQL statements) and the resulting relation.

Use the following relations to answer the questions below:

STUDENT

studentID	name	major
123	Bill	IT
234	Sue	CS
345	Tom	SE
456	Ann	BUS
567	Linda	IT
678	Tom	IT
789	Sue	LA

DEPT

dept	location	avgSAT	students
IT	Bldg 70	1250	1250
CS	Bldg 10	1234	700
SE	Bldg 9	1237	500
HIS	Bldg 13	1109	67
ART	Bldg 56	1189	70

1. Show the department name and number of students for all departments with greater than 500 students.

   STATEMENT: DEPT[dept, students] WHERE students > 500

   RELATION:

dept	location	avgSAT	students
IT	Bldg 70	1250	1250
CS	Bldg 10	1234	700

2. Get a list of all students who are majoring in CS or IT. List their student ID and name.

   STATEMENT: STUDENT[studentID, name] WHERE major=”CS” OR major=”IT”

   RELATION:

studentID	name
123	Bill
234	Sue
567	Linda
678	Tom

3. Get a list of all student names.

   STATEMENT: STUDENT[name]

   RELATION:

name
Bill
Sue
Tom
Ann
Linda

4. Perform a product of the two relations **(do not show the resulting relation for this question, instead answer how many rows would be in the result set)**?

   STATEMENT: STUDENT PRODUCT DEPT

   NUMBER OF TUPLES IN RESULT SET: 35

5. Show the studentID and name of students majoring in areas not listed in the department relation.

   STATEMENT: STUDENT[studentID, name] WHERE STUDENT[major] NOT IN DEPT[dept]

   RELATION:

studentID	name
456	Ann
789	Sue